

Remote Sensing



- *Definition*
- *Concept*
- *System*

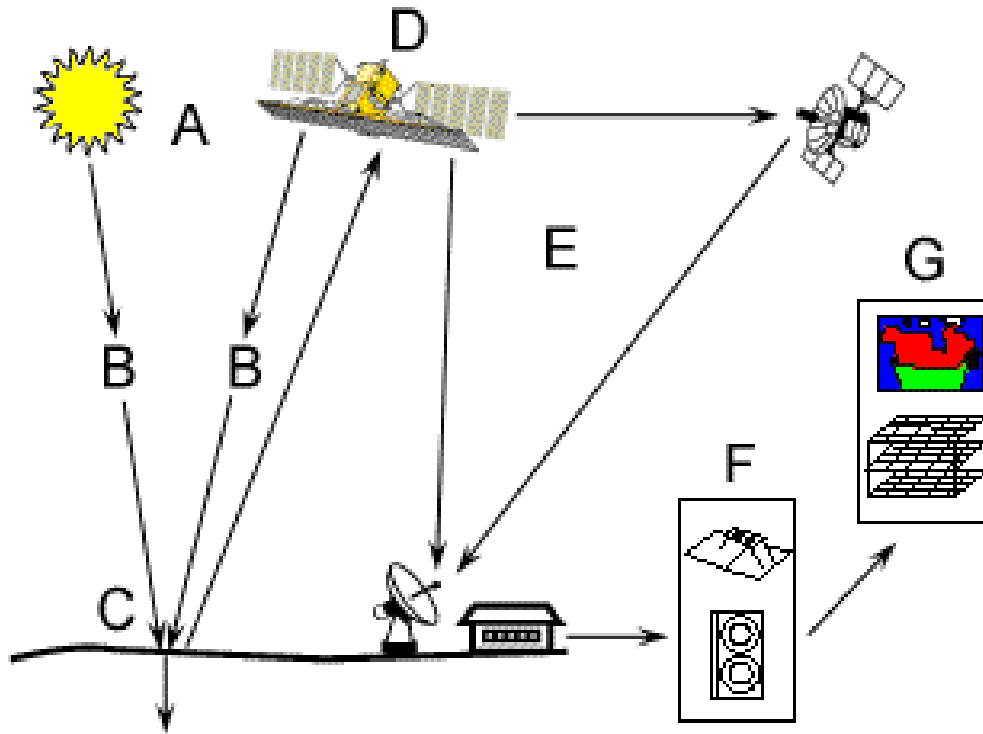
What is Remote Sensing?

*“Remote sensing is the science (and to some extent, art) of acquiring information about the Earth's surface without actually being in contact with it. **This is done by sensing and recording reflected or emitted energy and processing, analyzing, and applying that information.**”*

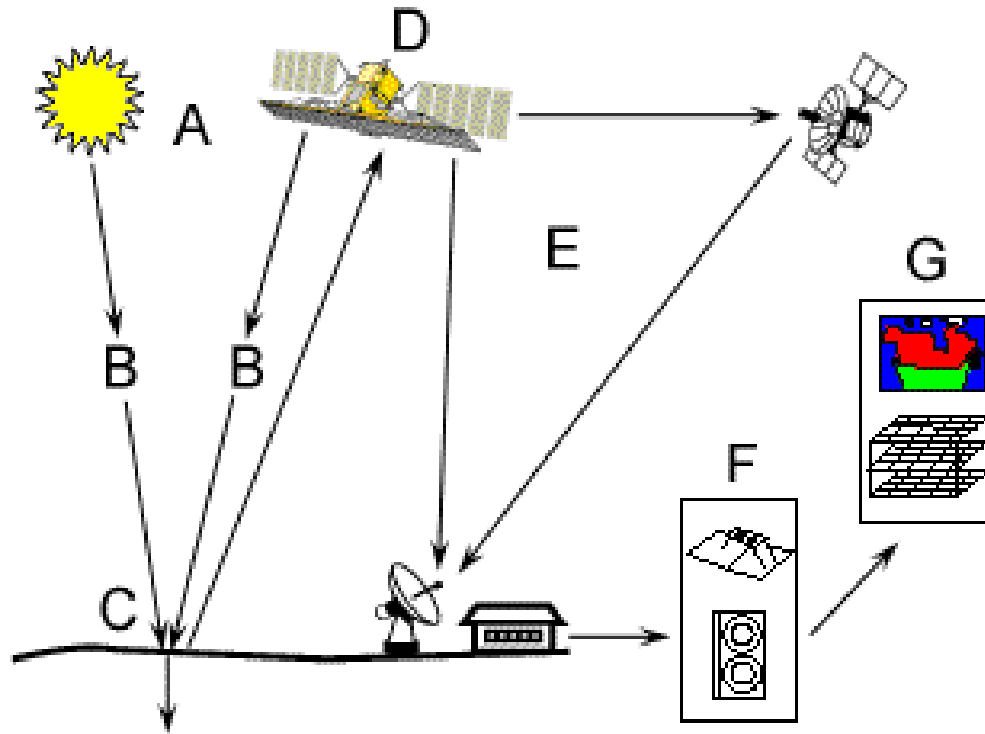
In much of remote sensing, **the process** involves an interaction between incident radiation and the targets of interest.

Remote imaging systems consist of **seven** elements.

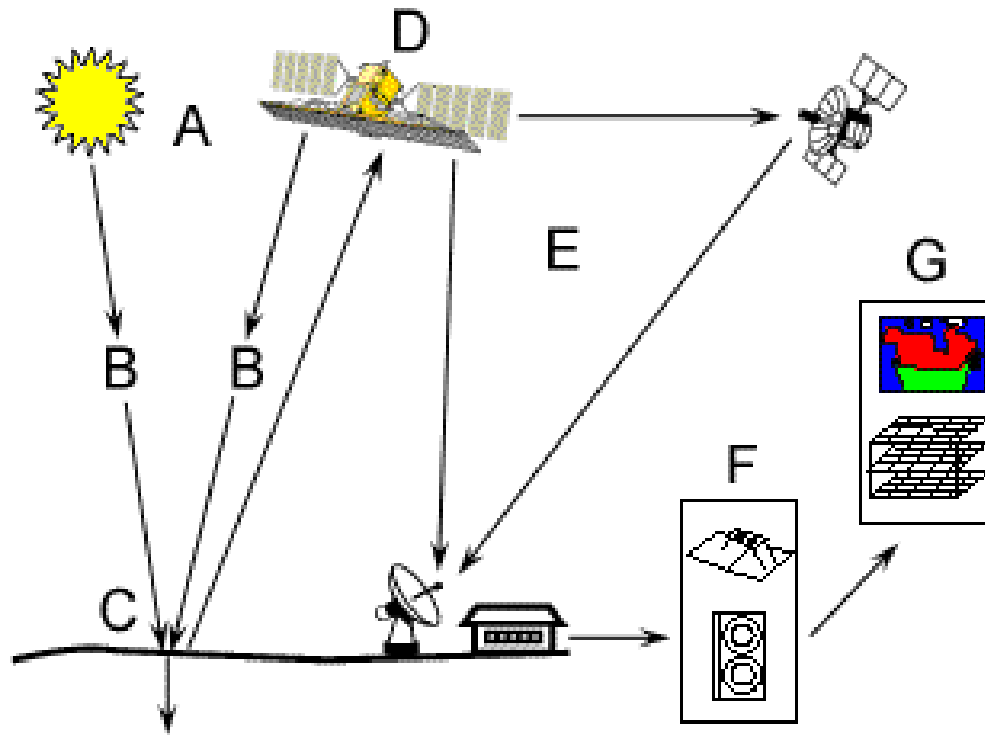
- However that remote sensing also involves the sensing of **emitted energy** and the use of **non-imaging sensors**.



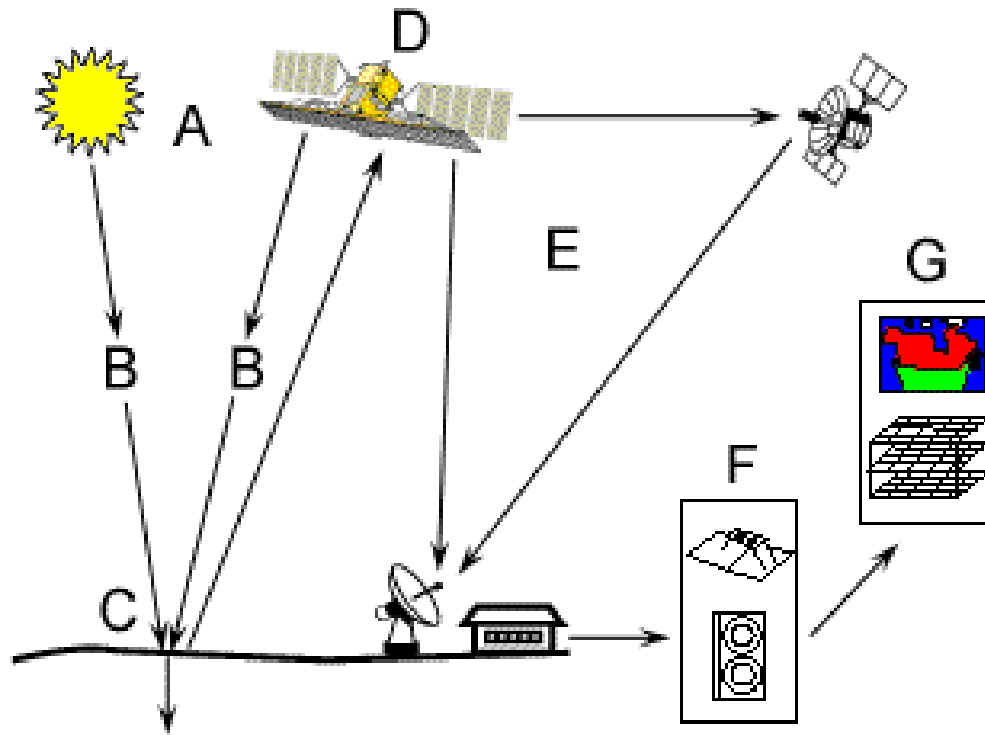
Energy Source or Illumination (A) – The first requirement for remote sensing is to have an energy source which illuminates or provides electromagnetic energy to the target of interest.



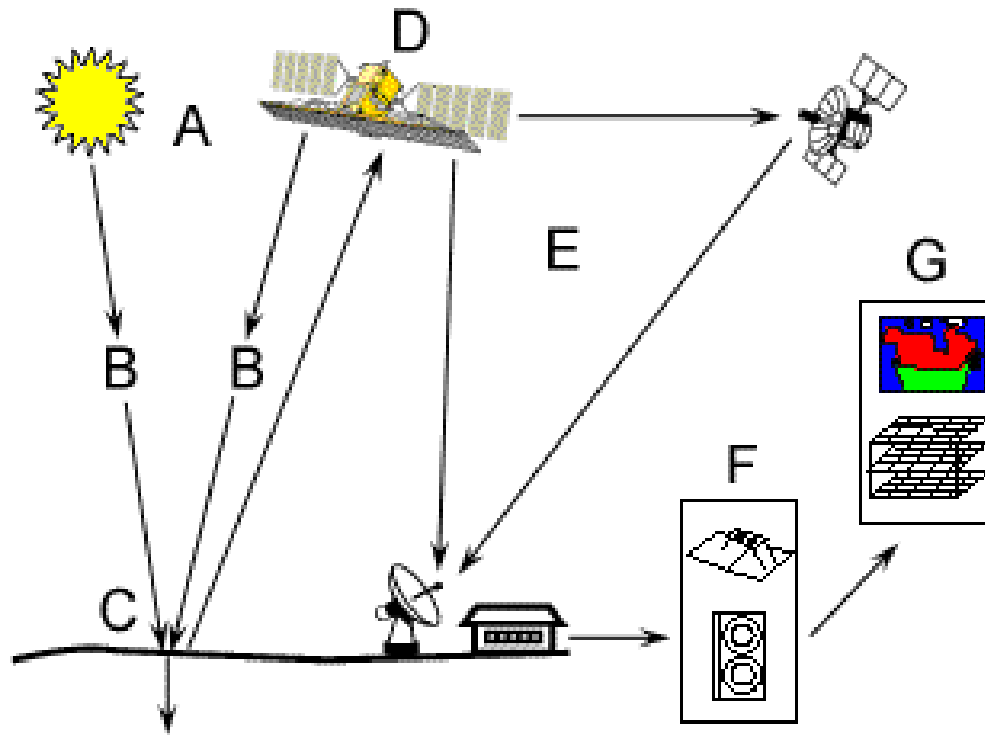
Radiation and the Atmosphere (B) – As the energy travels from its source to the target, it will come in contact with and interact with the atmosphere it passes through. This interaction may take place a second time as the energy travels from the target to the sensor.



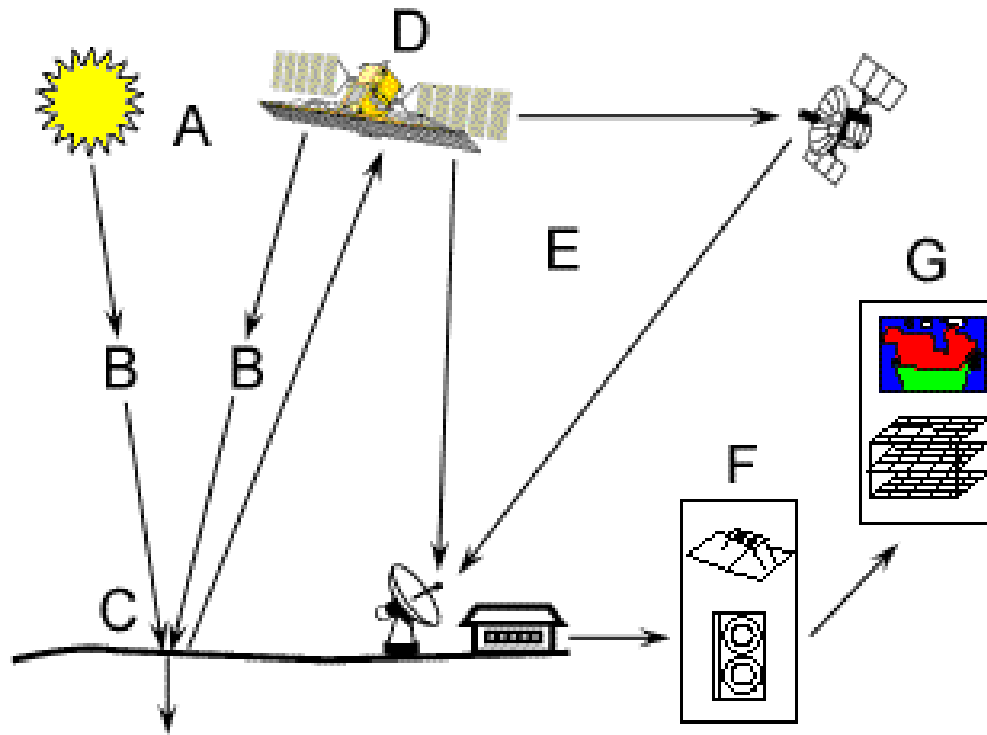
Interaction with the Target (C) - Once the energy makes its way to the target through the atmosphere, it interacts with the target depending on the properties of both the target and the radiation.



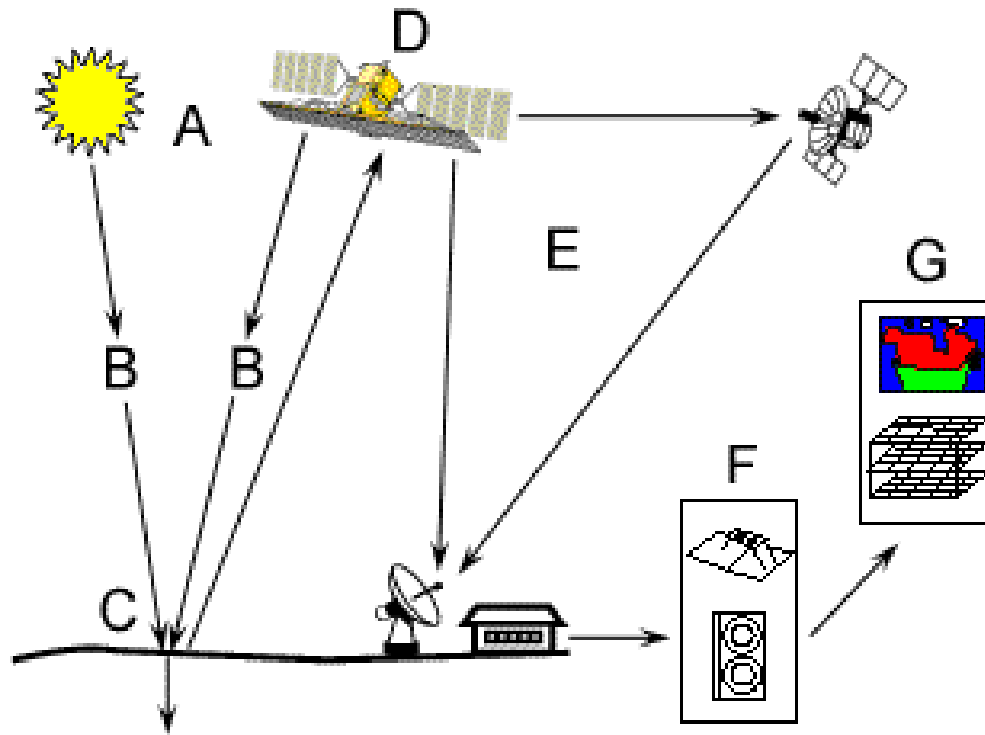
Recording of Energy by the Sensor (D) - After the energy has been scattered by, or emitted from the target, we require a sensor (remote - not in contact with the target) to collect and record the electromagnetic radiation.



Transmission, Reception, and Processing (E) -
The energy recorded by the sensor has to be transmitted, often in electronic form, to a receiving and processing station where the data are processed into an image (hardcopy and/or digital).



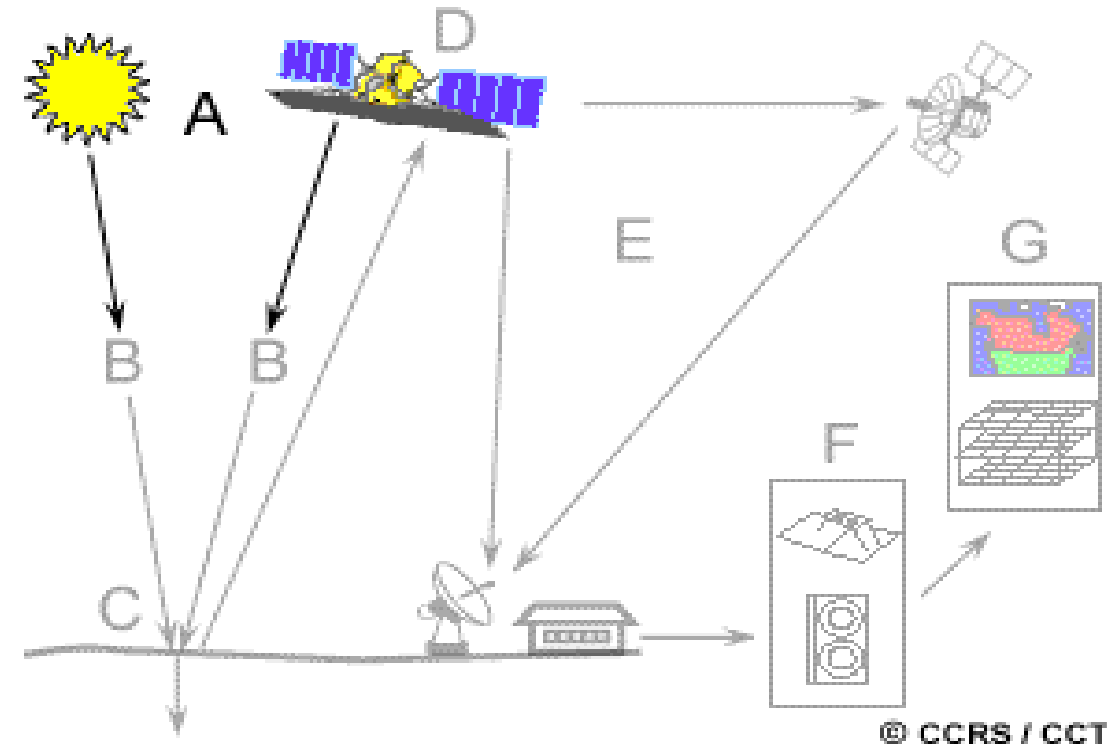
Interpretation and Analysis (F) - The processed image is interpreted, visually and/or digitally or electronically, to extract information about the target which was illuminated.



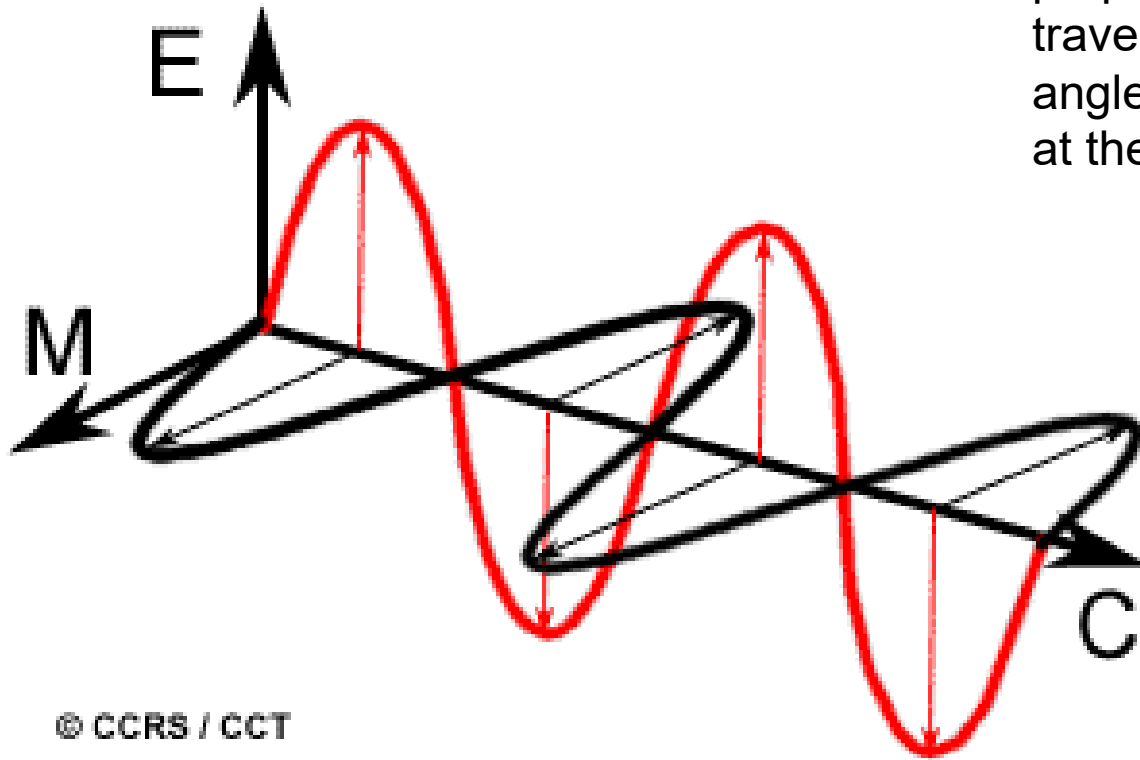
Application (G) - The final element of the remote sensing process is achieved when we apply the information we have been able to extract from the imagery about the target in order to better understand it, reveal some new information, or assist in solving a particular problem.

Electromagnetic Radiation

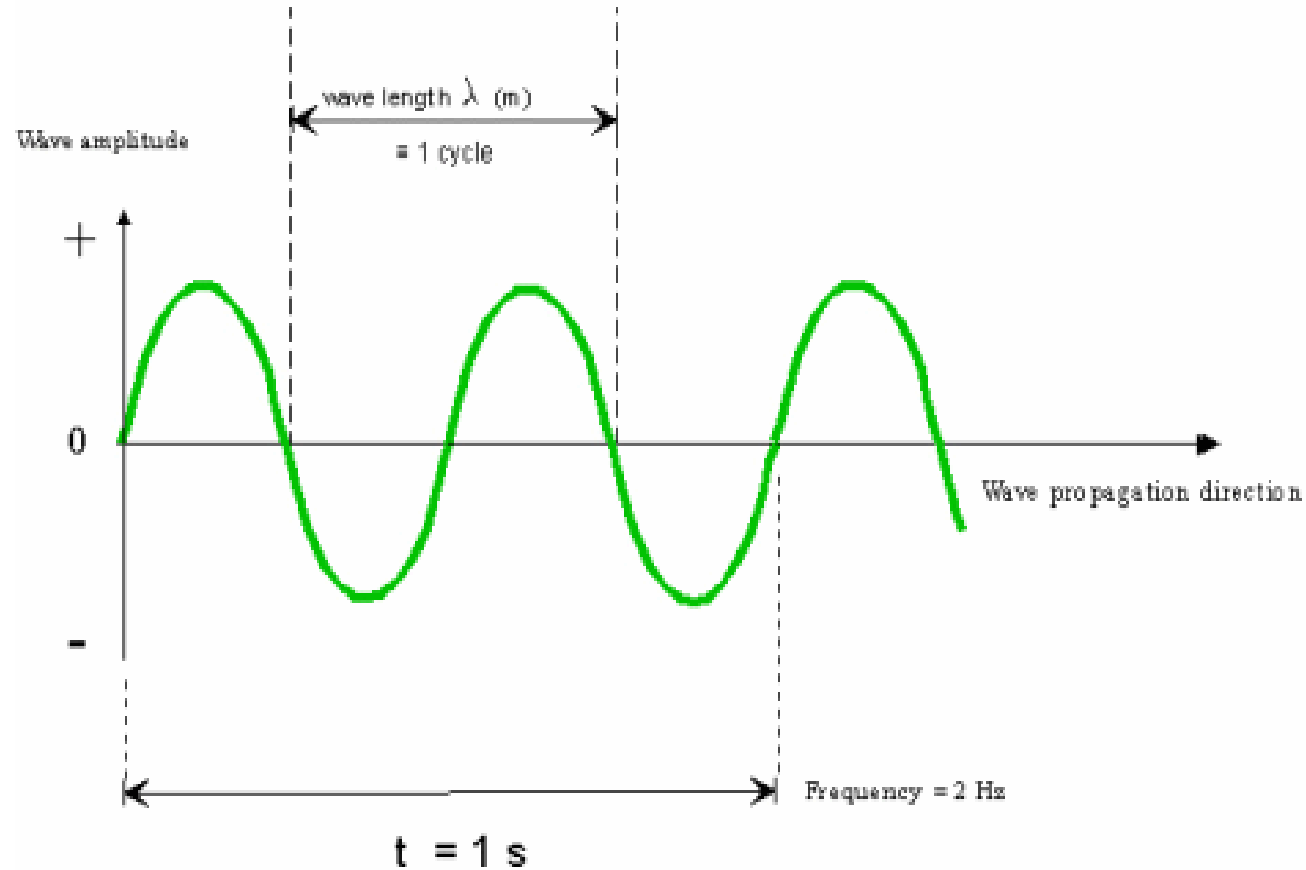
- ❑ The first requirement for remote sensing is to have an **energy source to illuminate the target** (unless the sensed energy is being emitted by the target).
- ❑ This energy is in the form of electromagnetic radiation.
- ❑ All electromagnetic radiation has fundamental properties and behaves in predictable ways according to the basics of wave theory.



Electromagnetic radiation consists of an electrical field (E) which varies in magnitude in a direction perpendicular to the direction in which the radiation is traveling, and a magnetic field (M) oriented at right angles to the electrical field. Both these fields travel at the speed of light (c).



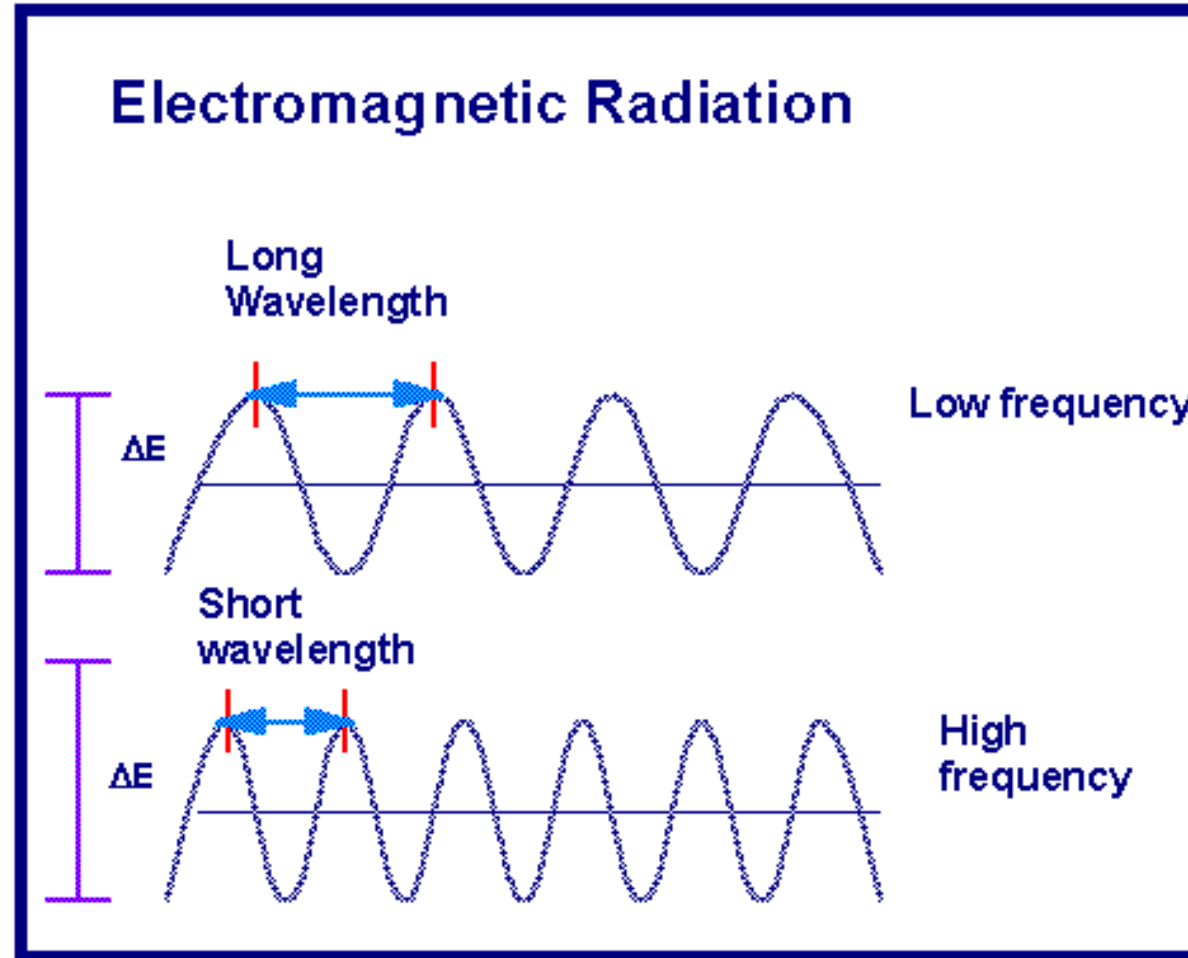
Two characteristics of electromagnetic radiation are important for understanding remote sensing, **wavelength** and **frequency**.

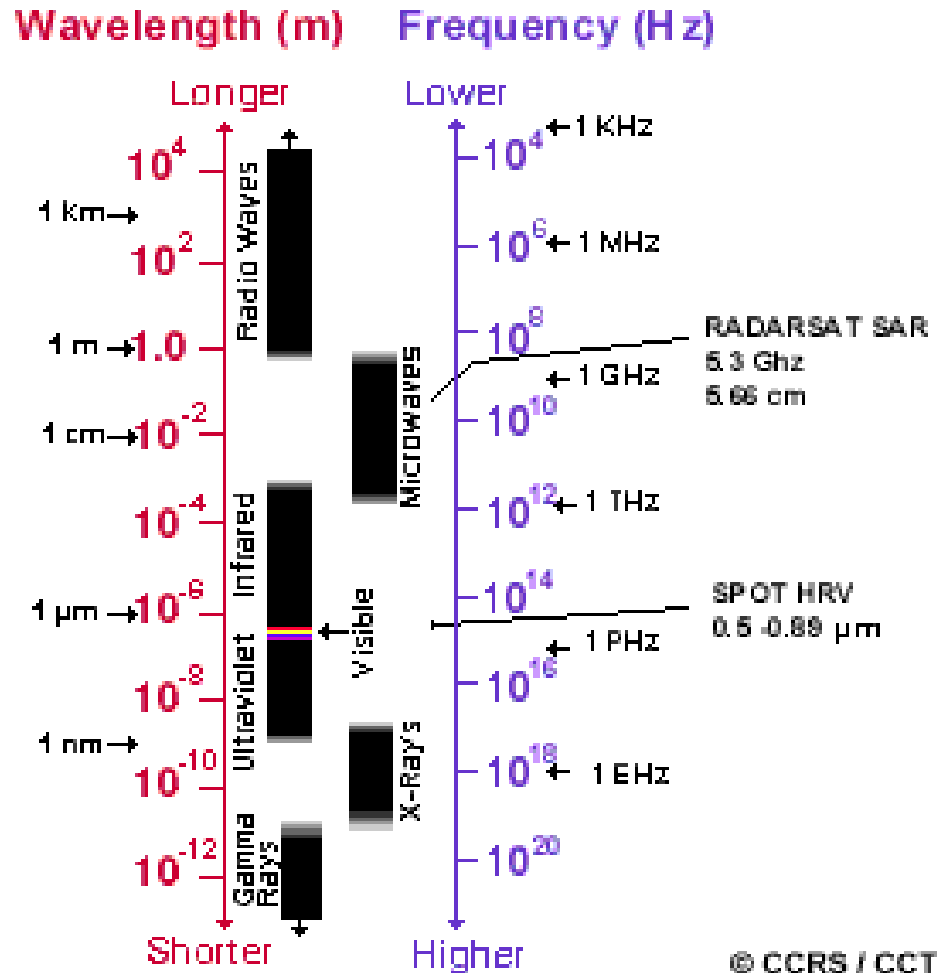


The wavelength is the length of one wave cycle, which can be measured as the distance between successive wave crests.

Frequency refers to the number of cycles of a wave passing a fixed point per unit of time.

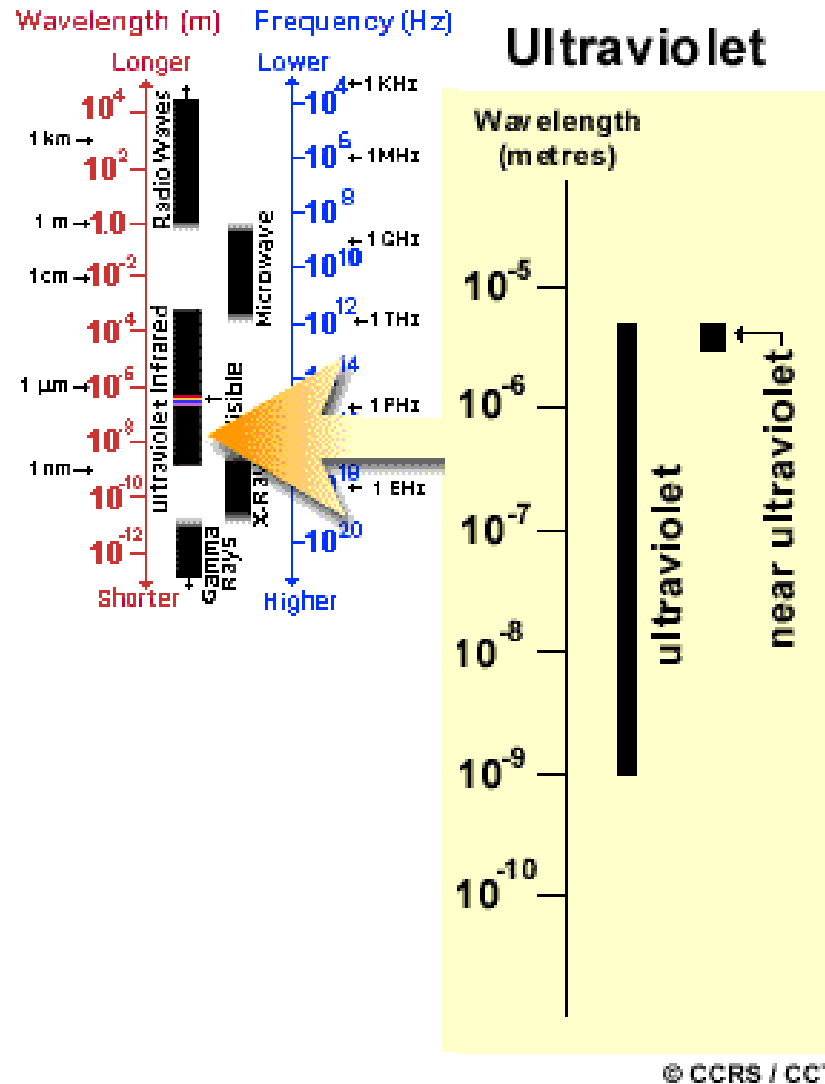
Relationship between wavelength and frequency



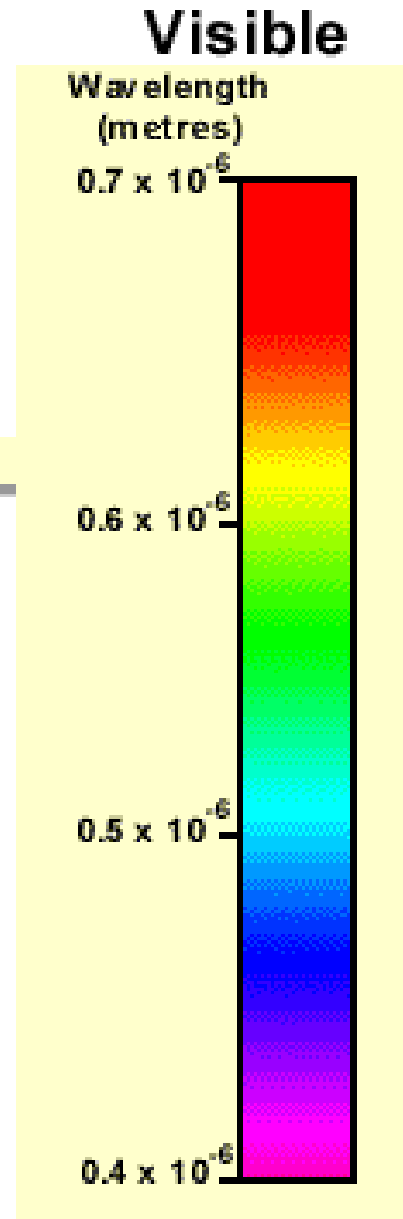
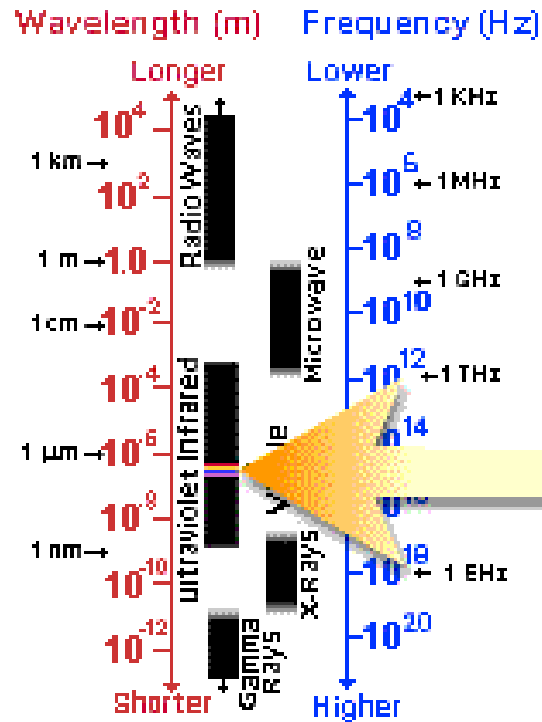


Electromagnetic Spectrum

The *electromagnetic spectrum* ranges from the shorter wavelengths (including gamma and x-rays) to the longer wavelengths (including microwaves and broadcast radio waves).



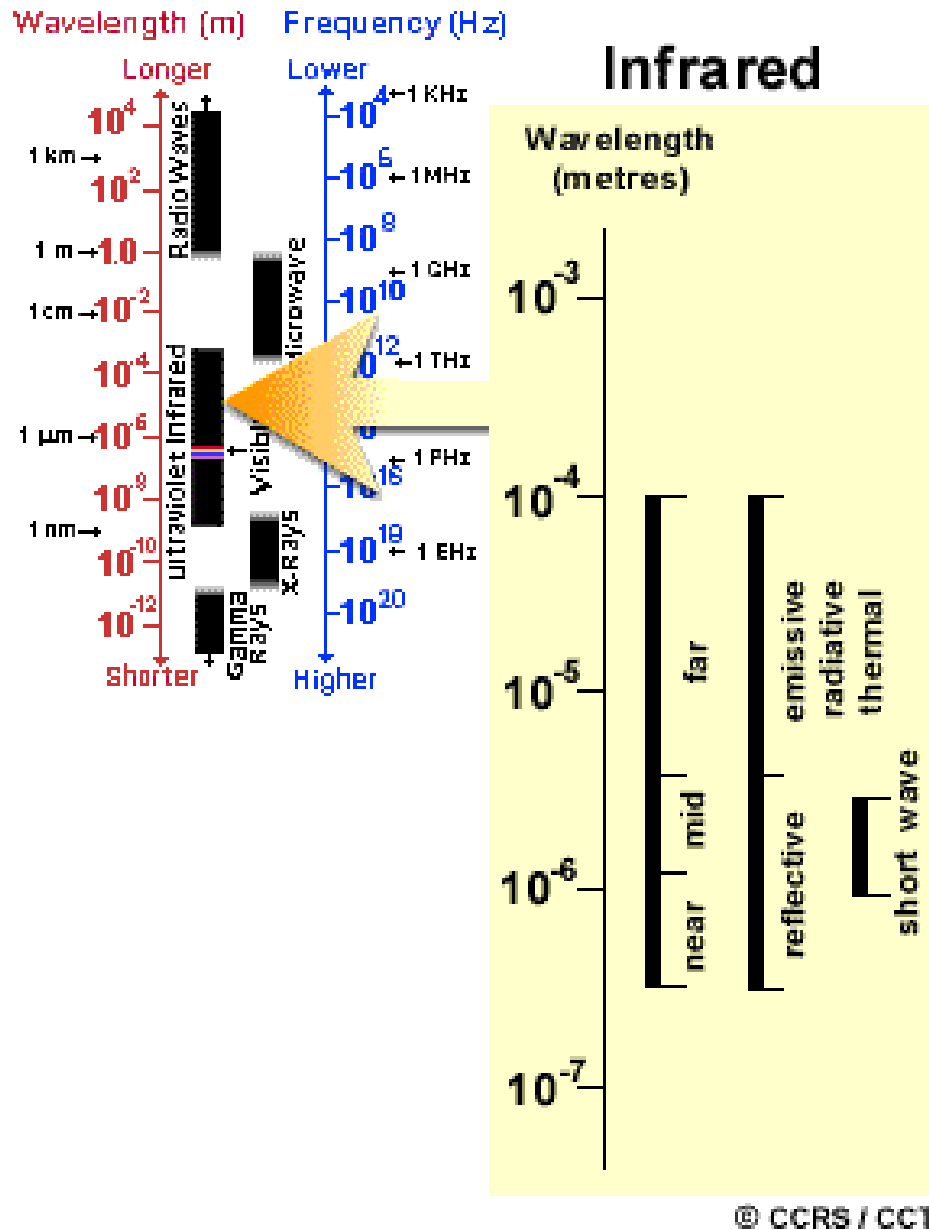
- For most purposes, the **ultraviolet or UV** portion of the spectrum has the shortest wavelengths which are practical for remote sensing.
- This radiation is just beyond the violet portion of the visible wavelengths, hence its name.
- Some Earth surface materials, primarily rocks and minerals, fluoresce or emit visible light when illuminated by UV radiation.



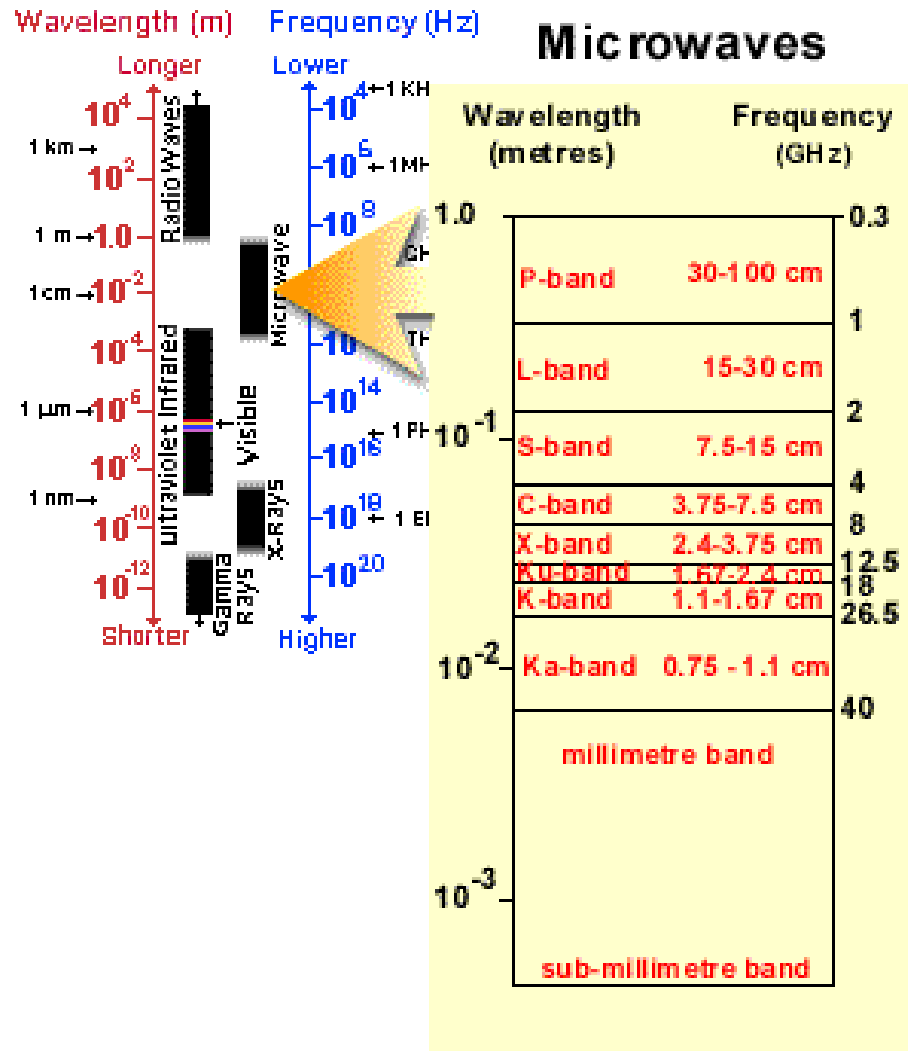
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Violet: 0.4 - 0.425 μ m
Indigo: 0.425-0.446 μ m
Blue: 0.446 - 0.500 μ m
Green: 0.500 - 0.578 μ m
Yellow: 0.578 - 0.592 μ m
Orange: 0.592 - 0.620 μ m
Red: 0.620 - 0.7 μ m

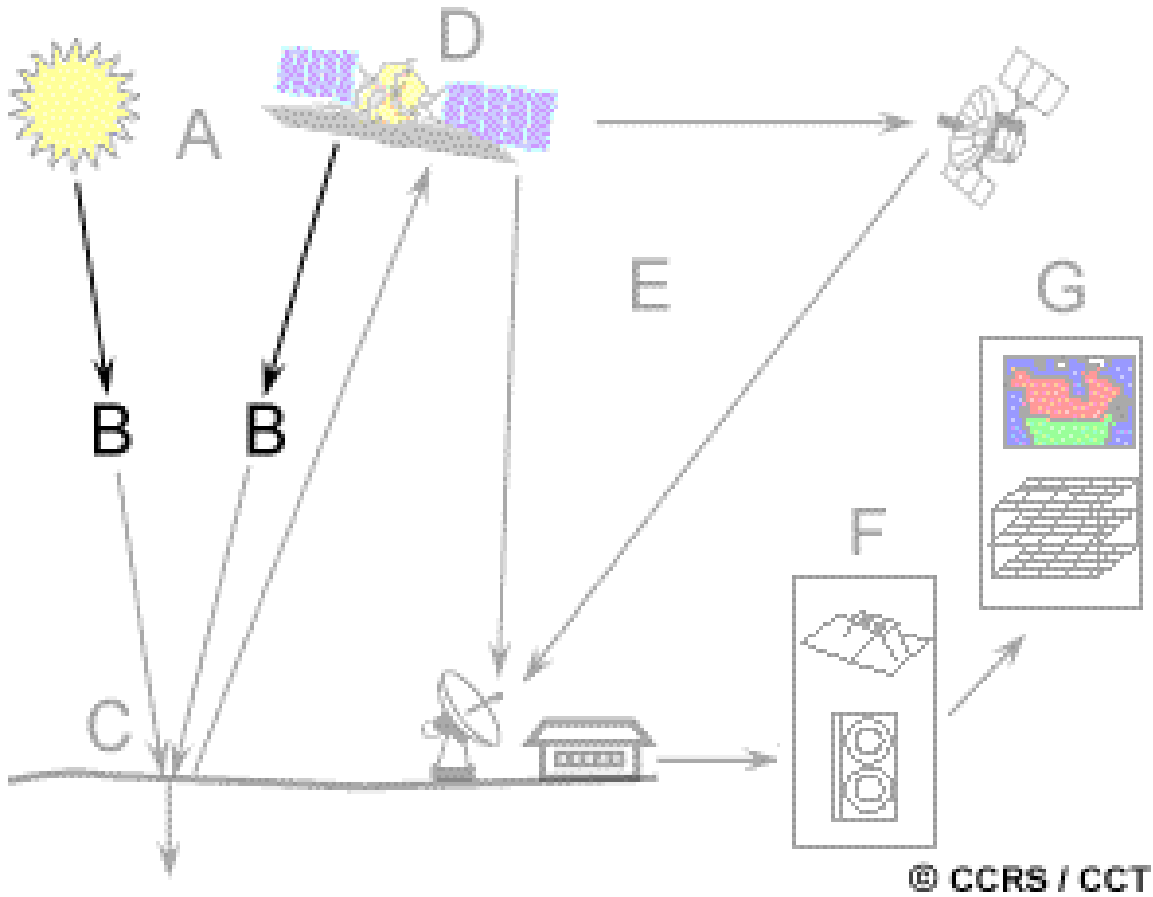
- The light which our eyes - our "remote sensors" - can detect is part of the **visible spectrum**.
- There is a lot of radiation around us which is "invisible" to our eyes, but can be detected by other remote sensing instruments and used to our advantage.
- The visible wavelengths cover a range from approximately 0.4 to 0.7 μ m.
- The longest visible wavelength is red and the shortest is violet.
- It is important to note that this is the only portion of the spectrum we can associate with the concept of **colours**.



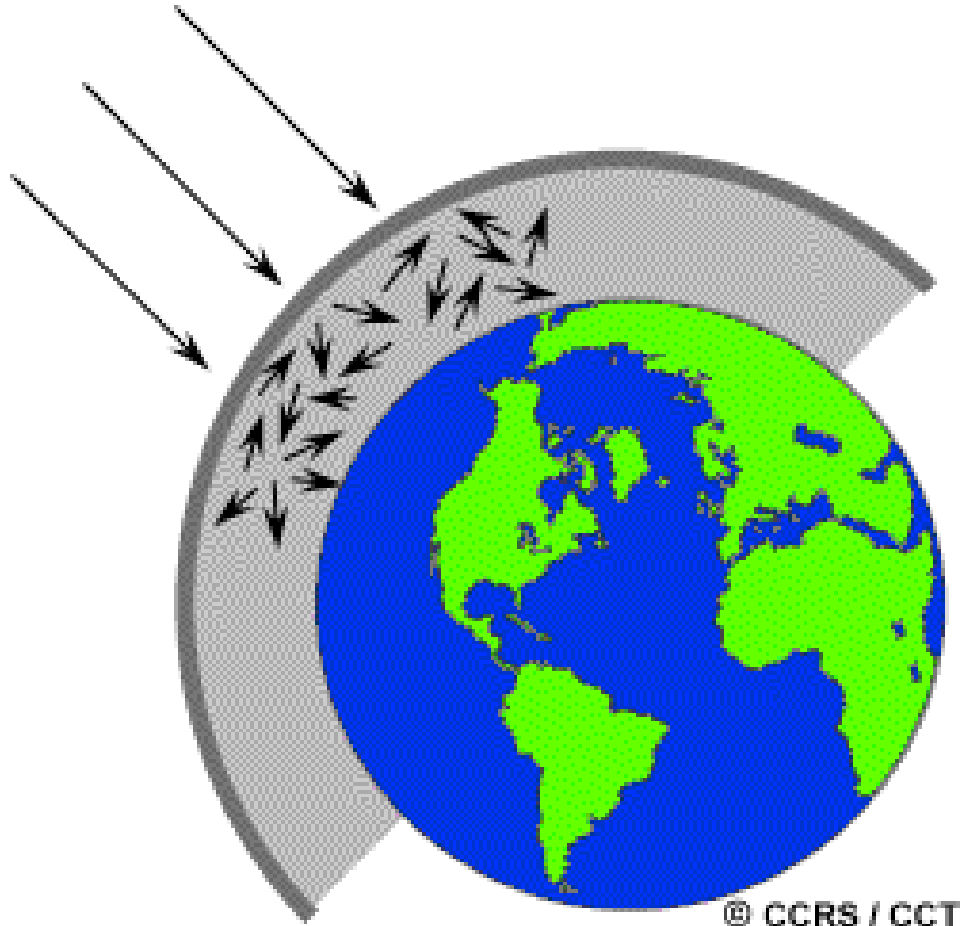
- ❑ Infrared (IR) region which covers the wavelength range from approximately $0.7 \mu\text{m}$ to $100 \mu\text{m}$ - more than 100 times as wide as the visible portion.
- ❑ The infrared region can be divided into two categories based on their radiation properties - the **reflected IR**, and the emitted or **thermal IR**.
- ❑ Radiation in the reflected IR region is used for remote sensing purposes in ways very similar to radiation in the visible portion.
- ❑ The reflected IR covers wavelengths from approximately $0.7 \mu\text{m}$ to $3.0 \mu\text{m}$.



- ❑ The portion of the spectrum of more recent interest to remote sensing is the **microwave region** from about 1 mm to 1 m.
- ❑ This covers the longest wavelengths used for remote sensing.
- ❑ The shorter wavelengths have properties similar to the thermal infrared region while the longer wavelengths approach the wavelengths used for radio broadcasts.
- ❑ Because of the special nature this spectrum is widely used in Canada.



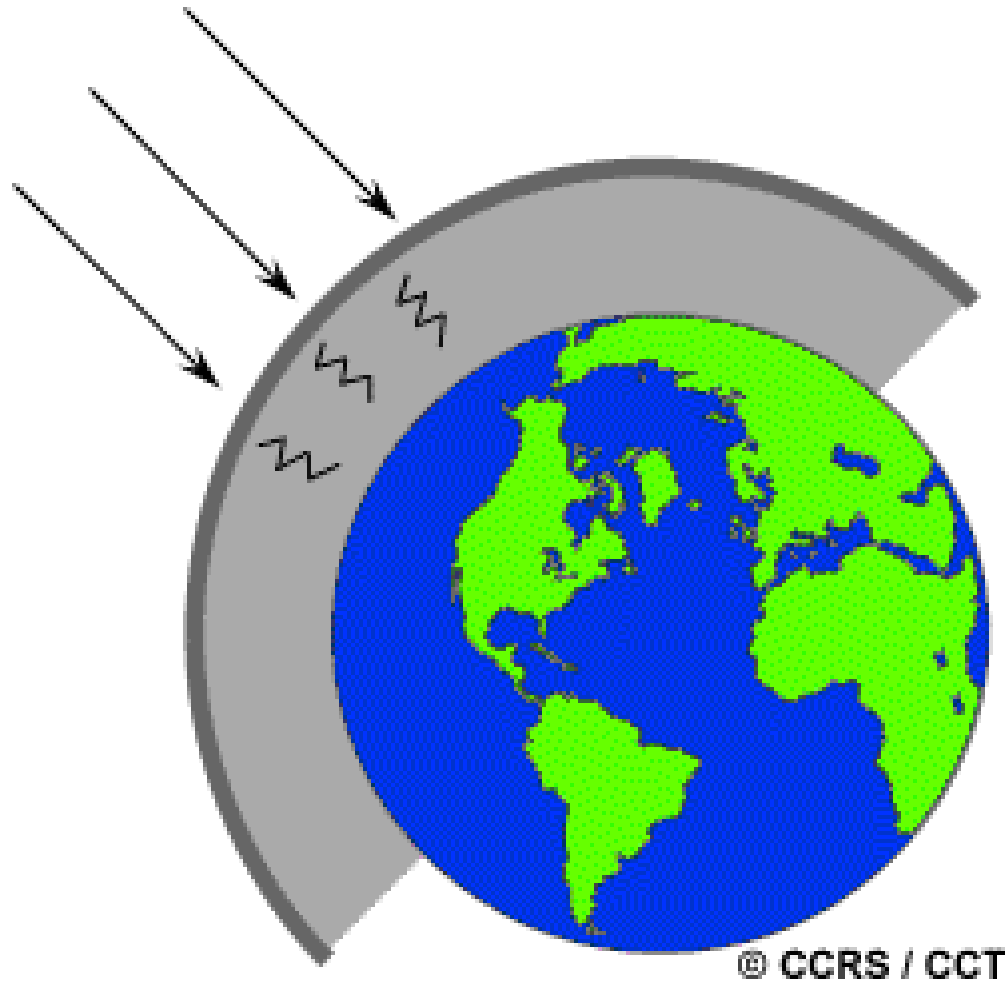
- ❑ Before radiation used for remote sensing reaches the Earth's surface it has to travel through some distance of the Earth's atmosphere.
- ❑ Particles and gases in the atmosphere can affect the incoming light and radiation. These effects are caused by the mechanisms of **scattering** and **absorption**.



- ❑ **Scattering** occurs when particles or large gas molecules present in the atmosphere interact with and cause the electromagnetic radiation to be redirected from its original path.
- ❑ How much scattering takes place depends on several factors including
 - the wavelength of the radiation,
 - the abundance of particles or gases, and
 - the distance the radiation travels through the atmosphere.

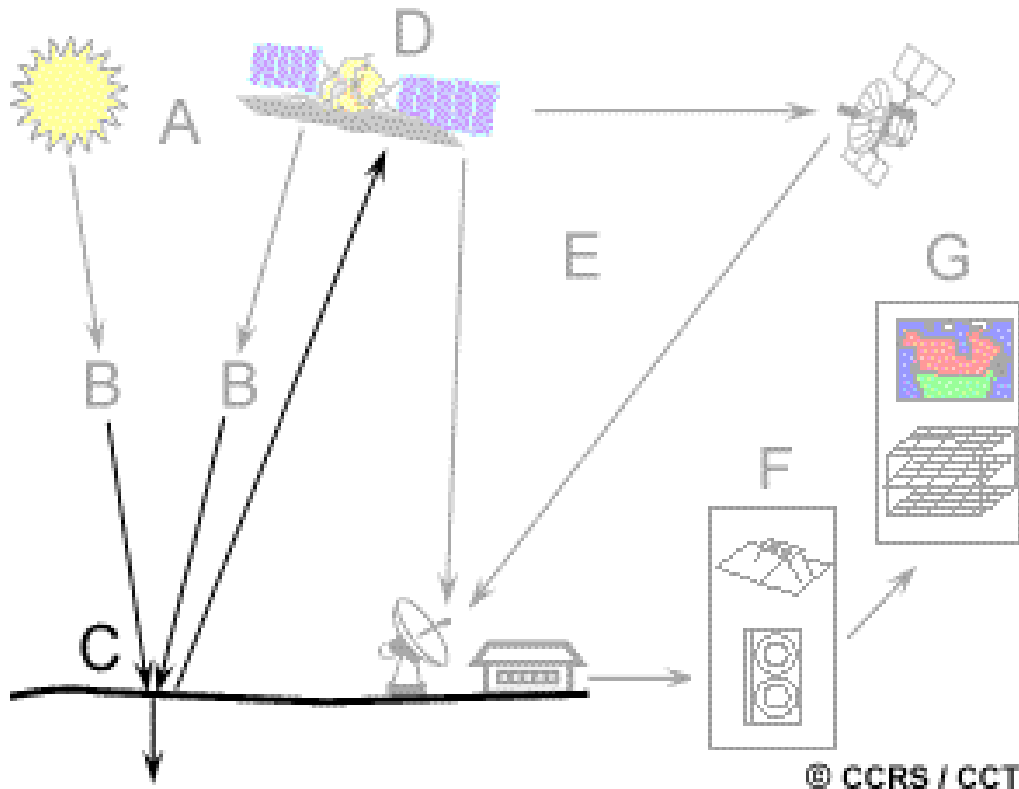
Types:

- **Rayleigh scattering**
- **Mie scattering**
- **Nonselective scattering**



- ❑ **Absorption** is the other main mechanism at work when electromagnetic radiation interacts with the atmosphere.
- ❑ In contrast to scattering, this phenomenon causes molecules in the atmosphere to absorb energy at various wavelengths.
- ❑ Ozone, carbon dioxide, and water vapour are the three main atmospheric constituents which absorb radiation.

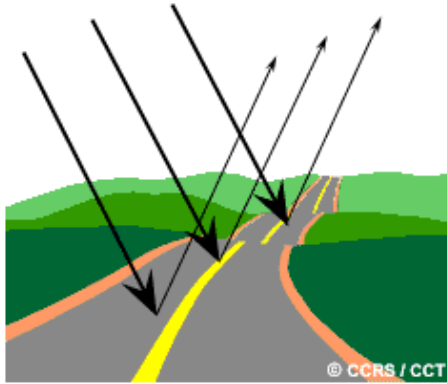
Radiation - Target Interactions



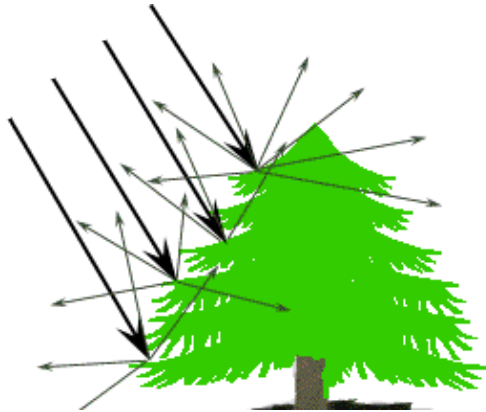
- ❑ Radiation that is not absorbed or scattered in the atmosphere can reach and interact with the Earth's surface.
- ❑ There are three (3) forms of interaction that can take place when energy strikes, or is **incident (I)** upon the surface.
- ❑ These are: **absorption (A)**; **transmission (T)**; and **reflection (R)**.
- ❑ The total incident energy will interact with the surface in one or more of these three ways.
- ❑ The proportions of each will depend on the wavelength of the energy and the material and condition of the feature.



- ❑ In remote sensing, we are most interested in measuring the radiation reflected from targets.
- ❑ We refer to two types of reflection, which represent the two extreme ends of the way in which energy is reflected from a target:
 - ❑ **specular reflection** and
 - ❑ **diffuse reflection.**

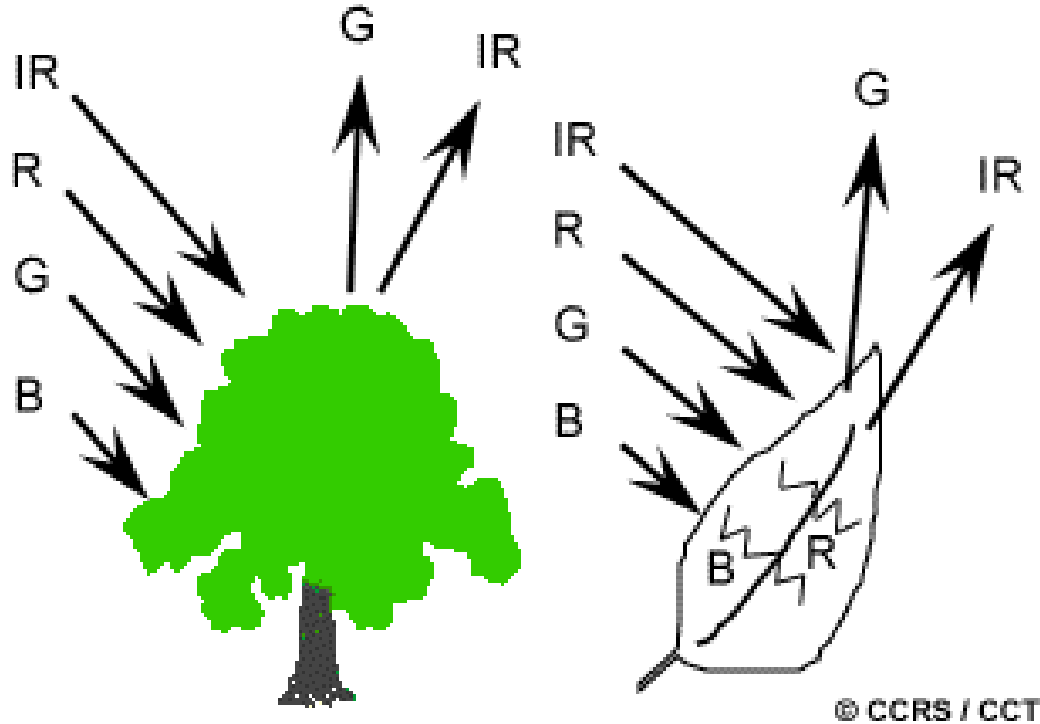


When a surface is smooth we get **specular** or mirror-like reflection where all (or almost all) of the energy is directed away from the surface in a single direction.

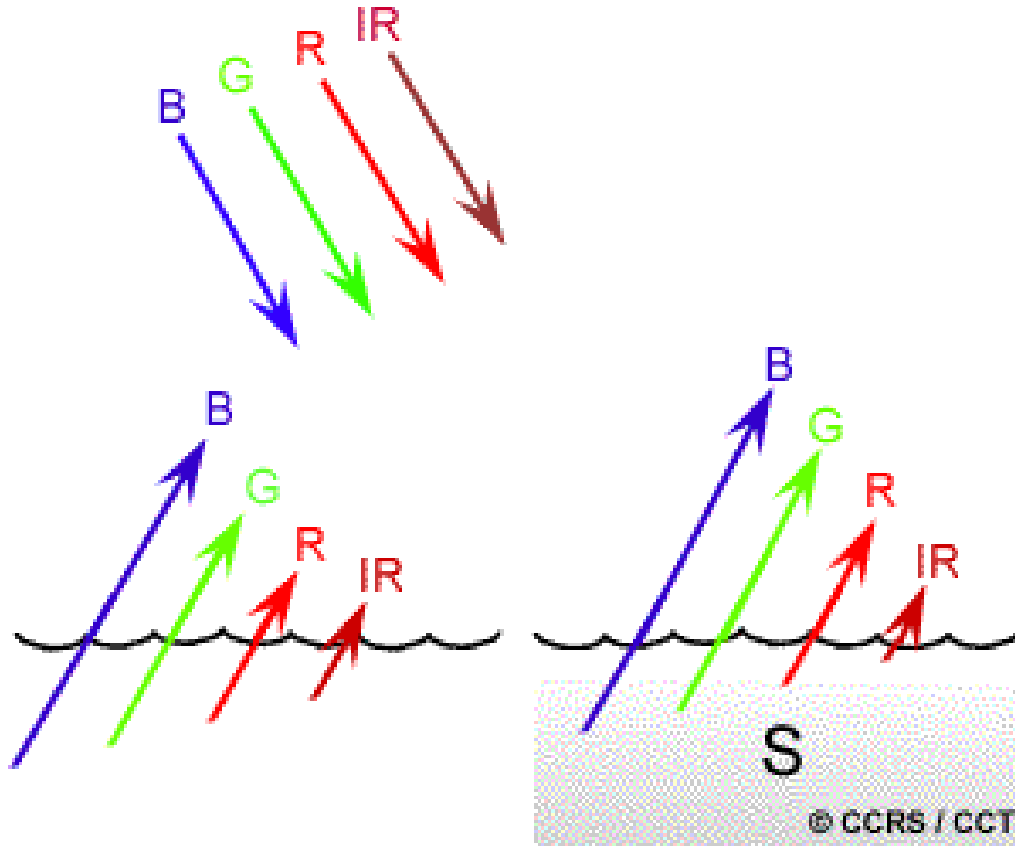


Diffuse reflection occurs when the surface is rough and the energy is reflected almost uniformly in all directions.

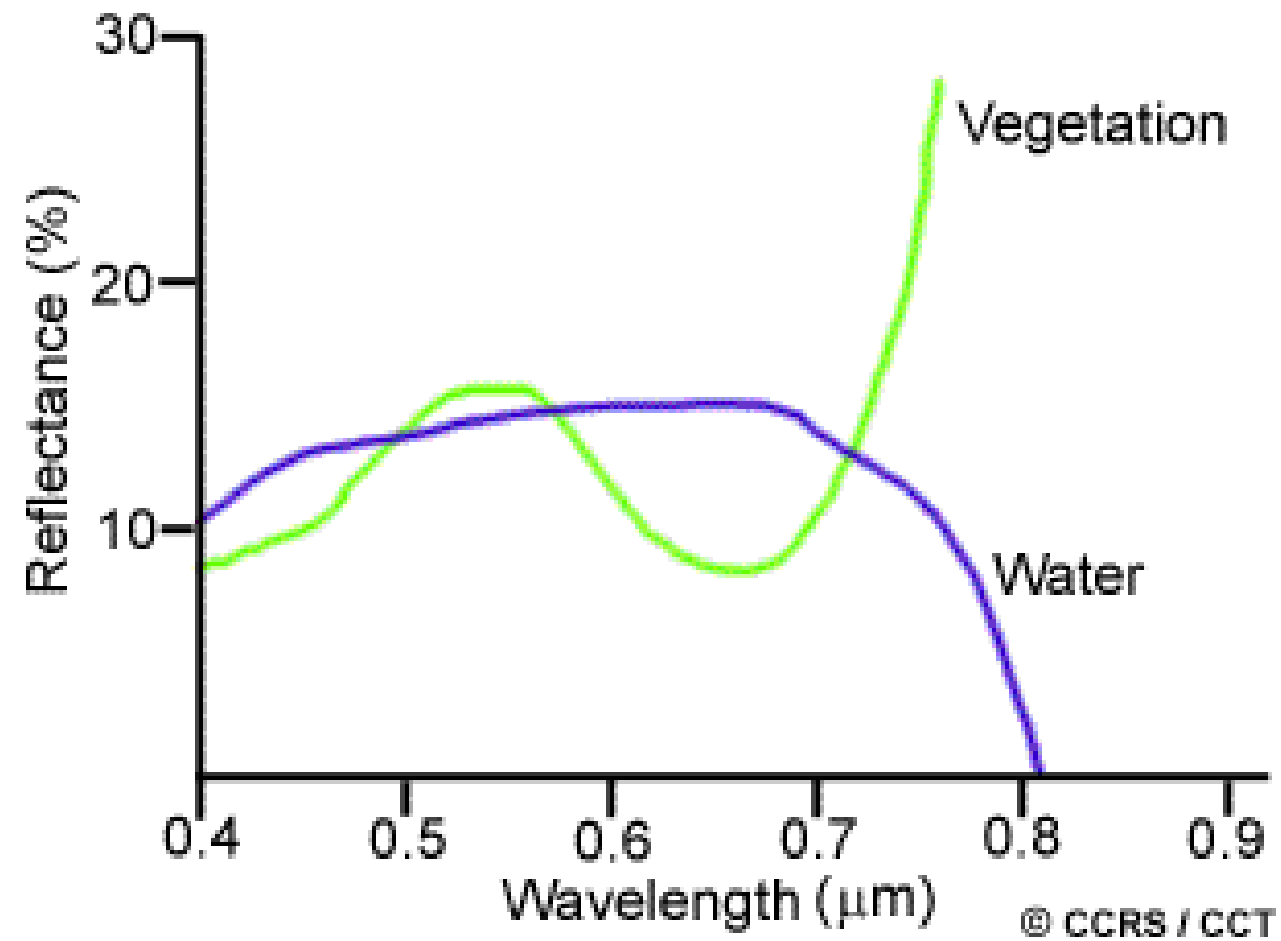
- ❑ Most earth surface features lie somewhere between perfectly specular or perfectly diffuse reflectors.
- ❑ Whether a particular target reflects specularly or diffusely, or somewhere in between, depends on the surface roughness of the feature in comparison to the wavelength of the incoming radiation.
- ❑ If the wavelengths are much smaller than the surface variations or the particle sizes that make up the surface, diffuse reflection will dominate.
- ❑ For example, fine grained sand would appear fairly smooth to long wavelength microwaves but would appear quite rough to the visible wavelengths.



- A chemical compound in leaves called chlorophyll strongly absorbs radiation in the red and blue wavelengths but reflects green wavelengths.
- Leaves appear "greenest" to us in the summer, when chlorophyll content is at its maximum.
- In autumn, there is less chlorophyll in the leaves, so there is less absorption and proportionately more reflection of the red wavelengths, making the leaves appear red or yellow (yellow is a combination of red and green wavelengths).

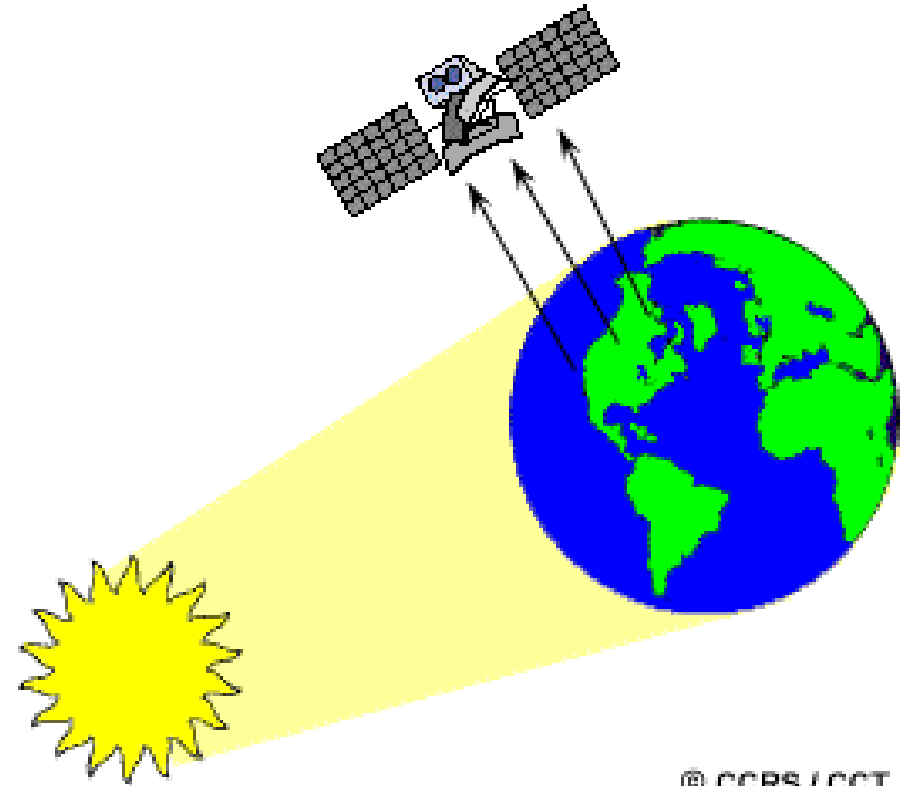


- Longer wavelength visible and near infrared radiation is absorbed more by water than shorter visible wavelengths. Thus water typically looks blue or blue-green due to stronger reflectance at these shorter wavelengths, and darker if viewed at red or near infrared wavelengths.
- If there is suspended sediment present in the upper layers of the water body, then this will allow better reflectivity and a brighter appearance of the water.
- The apparent colour of the water will show a slight shift to longer wavelengths.
- Suspended sediment (S) can be easily confused with shallow (but clear) water, since these two phenomena appear very similar.
- Chlorophyll in algae absorbs more of the blue wavelengths and reflects the green, making the water appear more green in colour when algae is present.



Passive vs. Active Sensors

- ☐ The sun provides a very convenient source of energy for remote sensing.
- ☐ The sun's energy is either **reflected**, as it is for visible wavelengths, or absorbed and then **reemitted**, as it is for thermal infrared wavelengths.
- ☐ Remote sensing systems which measure energy that is naturally available are called **passive sensors**.
- ☐ Passive sensors can only be used to detect energy when the naturally occurring energy is available.
- ☐ For all reflected energy, this can only take place during the time when the sun is illuminating the Earth.



- **Active sensors**, on the other hand, provide their own energy source for illumination. The sensor emits radiation which is directed toward the target to be investigated.
- The radiation reflected from that target is detected and measured by the sensor.

(+) Advantages for active sensors include the ability to obtain measurements anytime, regardless of the time of day or season.

(+) Active sensors can be used for examining wavelengths that are not sufficiently provided by the sun, such as microwaves, or to better control the way a target is illuminated.

- However, active systems require the generation of a fairly large amount of energy to adequately illuminate targets.
- Some examples of active sensors are a laser fluorosensor and a Synthetic Aperture Radar (SAR).

